Keshav Sharma

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EDUCATION

Vellore Institute of Technology **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) *Graduation: May 2027*

SKILL SUMMARY

Languages: Python, Java, C/C++, SQL, JavaScript, R
Software Engineering: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Database Management Systems, System Design, Distributed Systems, REST APIs, Multithreading & Concurrent Programming, Design Patterns
Backend & Frameworks: FastAPI, Flask, React.js, Node.js, REST APIs, SQLAlchemy
Databases & Storage: MySQL, SQLite, Redis, Database Design
Tools & Platforms: Git, GitHub, Docker, GitHub Actions, VS Code, Jupyter Noteboob
Developer Productivity: GitHub Copilot, Cursor, Claude, ChatGPT

EXPERIENCE

VOIS – Vodafone Idea Foundation* |LINK** ***Aug – Sep, 2025

- Analyzed 8,500+ real-world records using Julius AI and LLM-assisted workflows, extracting 15 actionable business insights through data exploration and prompt engineering.
- Developed structured analytical workflows, validated AI-generated outputs, and translated findings into 4 business recommendations presented to 3 mentors.
- Completed an industry-sponsored capstone project, earning certification from VOIS, Vodafone Idea Foundation, and AICTE.

PROJECTS

***RedisLite* |LINK**

- Built RedisLite, a Redis-inspired in-memory key-value database in Python featuring a custom RESP protocol parser, multithreaded TCP server, thread-safe storage engine, and O(1) LRU cache.
- Designed a modular four-layer architecture implementing Write-Ahead Logging (WAL), crash recovery, lazy and active TTL expiration, and concurrent request handling for reliable data persistence.
- Developed 46 automated unit tests, benchmarking 619K SET operations/sec in memory and 2.8K operations/sec with durable persistence while ensuring correctness and reliability.

***API Rate Limiter Library* |LINK**

- Developed a production-ready API rate-limiting library in Python implementing five industry-standard throttling algorithms with Redis integration and reusable middleware for Flask and FastAPI.
- Designed a modular architecture supporting concurrent request handling, storage abstraction, and framework-independent integration for scalable backend applications.
- Built 30+ automated unit tests and published the library to PyPI for public installation and reuse.

***AI-Powered Expense Receipt Scanner & Analytics Platform* |LINK**

- Developed a full-stack expense management platform using FastAPI, React, Vite, SQLAlchemy, and SQLite, automating receipt processing and reducing manual processing time by 40% using EasyOCR, OpenCV, and NLP-based data extraction.
- Designed and deployed secure backend services with JWT Authentication, Google OAuth, RESTful APIs, and complete CRUD functionality for receipt and expense management, enabling scalable and maintainable application architecture.
- Built an interactive analytics dashboard featuring budget tracking, category-wise spending insights, Excel export, and responsive UI, deploying the application on Render and Vercel for production access.

Certifications

- **Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |LINK** **2026**
  Demonstrated expertise in Generative AI, LLMs, Prompt Engineering, RAG, AI Agents, Agentic Workflows, Responsible AI, and OCI AI services.
- **Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate |LINK** **2026**
  Demonstrated foundational knowledge of AI, Machine Learning, Generative AI, LLMs, Prompt Engineering, and Oracle Cloud AI services.